

Math 363/663 Homework 10 Solutions

Problem 1. Find the Fourier integrals of the functions

$$f(x) = \begin{cases} -|x| + 1 & |x| \leq 1 \\ 0 & |x| > 1 \end{cases} \quad \text{and} \quad g(x) = \begin{cases} 1 & 0 < x < 1 \\ -1 & -1 < x < 0 \\ 0 & |x| > 1. \end{cases}$$

For what values of x do the equalities

$$\text{FI}(f)(x) = f(x) \quad \text{and} \quad \text{FI}(g)(x) = g(x)$$

hold?

For f ,

$$\begin{aligned} A(\omega) &= \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \cos(\omega x) dx = \frac{1}{\pi} \int_{-1}^1 (-|x| + 1) \cos(\omega x) dx \\ &= \frac{2}{\pi} \int_0^1 (-x + 1) \cos(\omega x) dx \quad (\text{the integrand is even}) \\ &= \frac{2}{\pi} \left[(-x + 1) \frac{\sin(\omega x)}{\omega} - \frac{\cos(\omega x)}{\omega^2} \right] \Big|_0^1 \quad (\text{integration by parts}) \\ &= \frac{2}{\pi} \frac{1 - \cos \omega}{\omega^2}, \end{aligned}$$

and $B(\omega) = 0$ since f is even. Thus,

$$\text{FI}(f)(x) = \int_0^{\infty} A(\omega) \cos(\omega x) d\omega = \frac{2}{\pi} \int_0^{\infty} \frac{1 - \cos \omega}{\omega^2} \cos(\omega x) d\omega.$$

For g , $A(\omega) = 0$ since g is odd, and

$$\begin{aligned} B(\omega) &= \frac{1}{\pi} \int_{-\infty}^{\infty} g(x) \sin(\omega x) dx = \frac{2}{\pi} \int_0^{\infty} g(x) \sin(\omega x) dx \quad (\text{the integrand is even}) \\ &= \frac{2}{\pi} \int_0^1 \sin(\omega x) dx = -\frac{2}{\pi} \frac{\cos(\omega x)}{\omega} \Big|_0^1 \\ &= \frac{2}{\pi} \frac{1 - \cos \omega}{\omega}. \end{aligned}$$

Thus,

$$\text{FI}(g)(x) = \int_0^{\infty} B(\omega) \sin(\omega x) d\omega = \frac{2}{\pi} \int_0^{\infty} \frac{1 - \cos \omega}{\omega} \sin(\omega x) d\omega.$$

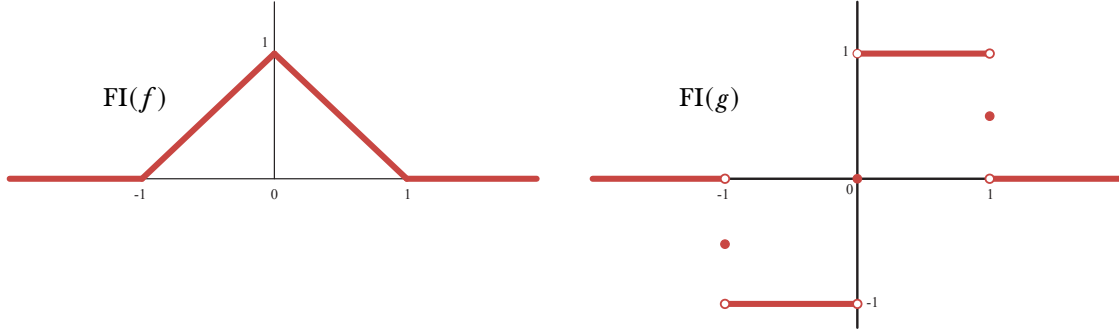
Since both f, g are piecewise C^1 on every bounded interval, the basic convergence theorem applies to them. Thus,

$$\text{FI}(f)(x) = \frac{1}{2}(f(x^+) + f(x^-)) = f(x)$$

for every $x \in \mathbb{R}$ since f is continuous everywhere, and

$$\text{FI}(g)(x) = \frac{1}{2}(g(x^+) + g(x^-)) = \begin{cases} g(x) & x \notin \{-1, 0, 1\} \\ -\frac{1}{2} & x = -1 \\ 0 & x = 0 \\ \frac{1}{2} & x = 1. \end{cases}$$

The graphs of $\text{FI}(f)$ and $\text{FI}(g)$ are shown below:



Problem 2. Use the result of problem 1 to compute the improper integrals

$$\int_0^\infty \frac{1 - \cos \omega}{\omega^2} d\omega \quad \text{and} \quad \int_0^\infty \frac{(1 - \cos \omega) \sin \omega}{\omega} d\omega.$$

Using the formulas for $\text{FI}(f)$ and $\text{FI}(g)$, we compute

$$1 = \text{FI}(f)(0) = \frac{2}{\pi} \int_0^\infty \frac{1 - \cos \omega}{\omega^2} d\omega \implies \int_0^\infty \frac{1 - \cos \omega}{\omega^2} d\omega = \frac{\pi}{2},$$

and

$$\frac{1}{2} = \text{FI}(g)(1) = \frac{2}{\pi} \int_0^\infty \frac{(1 - \cos \omega) \sin \omega}{\omega} d\omega \implies \int_0^\infty \frac{(1 - \cos \omega) \sin \omega}{\omega} d\omega = \frac{\pi}{4}.$$

Problem 3. Let $a > 0$ be a constant.

(i) Show that the formulas

$$e^{-ax} = \frac{2}{\pi} \int_0^\infty \frac{a}{a^2 + \omega^2} \cos(\omega x) d\omega \tag{1}$$

$$e^{-ax} = \frac{2}{\pi} \int_0^\infty \frac{\omega}{a^2 + \omega^2} \sin(\omega x) d\omega \tag{2}$$

hold for all $x > 0$.

Let $f(x) = e^{-ax}$ for $x > 0$. Let $f_e(x) = e^{-a|x|}$ be the even extension of f . Then

$$\text{FI}(f_e)(x) = \int_0^\infty A(\omega) \cos(\omega x) d\omega,$$

where

$$\begin{aligned}
 A(\omega) &= \frac{1}{\pi} \int_{-\infty}^{\infty} f_e(x) \cos(\omega x) dx = \frac{2}{\pi} \int_0^{\infty} f(x) \cos(\omega x) dx \\
 &= \frac{2}{\pi} \int_0^{\infty} e^{-ax} \cos(\omega x) dx \\
 &= \frac{2}{\pi} \frac{e^{-ax}}{a^2 + \omega^2} (-a \cos(\omega x) + \omega \sin(\omega x)) \Big|_0^{\infty} = \frac{2}{\pi} \frac{a}{a^2 + \omega^2}.
 \end{aligned}$$

Since f_e is piecewise C^1 on every bounded interval *and* continuous, the basic convergence theorem implies that $\text{FI}(f_e)(x) = f_e(x)$ for all $x \in \mathbb{R}$. In particular, for $x > 0$,

$$f(x) = \text{FI}(f_e)(x) = \frac{2}{\pi} \int_0^{\infty} \frac{a}{a^2 + \omega^2} \cos(\omega x) d\omega,$$

which proves (1). Similarly, let f_o be the odd extension of f :

$$f_o(x) = \begin{cases} -e^{ax} & x < 0 \\ 0 & x = 0 \\ e^{-ax} & x > 0. \end{cases}$$

Then

$$\text{FI}(f_o)(x) = \int_0^{\infty} B(\omega) \sin(\omega x) d\omega,$$

where

$$\begin{aligned}
 B(\omega) &= \frac{1}{\pi} \int_{-\infty}^{\infty} f_o(x) \sin(\omega x) dx = \frac{2}{\pi} \int_0^{\infty} f_o(x) \sin(\omega x) dx \\
 &= \frac{2}{\pi} \int_0^{\infty} e^{-ax} \sin(\omega x) dx \\
 &= \frac{2}{\pi} \frac{e^{-ax}}{a^2 + \omega^2} (-\omega \cos(\omega x) - a \sin(\omega x)) \Big|_0^{\infty} = \frac{2}{\pi} \frac{\omega}{a^2 + \omega^2}.
 \end{aligned}$$

Since f_o is piecewise C^1 on every bounded interval, the basic convergence theorem implies that $\text{FI}(f_o)(x) = (f_o(x^+) + f_o(x^-))/2$ for all $x \in \mathbb{R}$. Since $f_o(x^+) = f_o(x^-) = f(x)$ for $x > 0$, it follows that

$$f(x) = \text{FI}(f_o)(x) = \frac{2}{\pi} \int_0^{\infty} \frac{\omega}{a^2 + \omega^2} \sin(\omega x) d\omega,$$

which proves (2).

- (ii) Take d/dx of each side of the first formula, assuming differentiation under the integral sign is legitimate. What do you notice? What happens when you attempt a similar differentiation on the second formula?

Taking d/dx of each side of the formula (1), assuming differentiation under the integral sign is legitimate, gives

$$-ae^{-ax} = \frac{2}{\pi} \int_0^\infty \frac{-a\omega}{a^2 + \omega^2} \sin(\omega x) d\omega,$$

which, after canceling $-a$ from each side, is precisely the formula (2).

Let us now attempt the same differentiation on (2):

$$-ae^{-ax} = \frac{2}{\pi} \int_0^\infty \frac{\omega^2}{a^2 + \omega^2} \cos(\omega x) d\omega,$$

which is pure nonsense because the improper integral on the right is not even convergent. Intuitively, this is because for large ω the integrand behaves like $\cos(\omega x)$, which is not integrable on $[0, +\infty)$. To make this rigorous, fix $x > 0$ and note that since $\lim_{\omega \rightarrow +\infty} \omega^2/(a^2 + \omega^2) = 1$, we must have $\omega^2/(a^2 + \omega^2) > 1/2$ for all sufficiently large ω . Let

$$I(R) = \int_0^R \frac{\omega^2}{a^2 + \omega^2} \cos(\omega x) d\omega.$$

If $\int_0^\infty \omega^2 \cos(\omega x)/(a^2 + \omega^2) d\omega = \lim_{R \rightarrow +\infty} I(R)$ existed, we would have

$$\lim_{R \rightarrow +\infty} \left(I\left(R + \frac{\pi}{x}\right) - I(R) \right) = 0.$$

But, choosing $R_n = (4n - 1)\pi/(2x)$ for large integers n and noting that

$$\cos(\omega x) \geq 0 \quad \text{for} \quad \omega \in \left[R_n, R_n + \frac{\pi}{x} \right] = \left[\frac{(4n - 1)\pi}{2x}, \frac{(4n + 1)\pi}{2x} \right],$$

we see that

$$\begin{aligned} I\left(R_n + \frac{\pi}{x}\right) - I(R_n) &= \int_{R_n}^{R_n + \pi/x} \frac{\omega^2}{a^2 + \omega^2} \cos(\omega x) d\omega \\ &\geq \frac{1}{2} \int_{R_n}^{R_n + \pi/x} \cos(\omega x) d\omega = \frac{1}{x}, \end{aligned}$$

which shows $I(R_n + \pi/x) - I(R_n)$ does not tend to 0 as $n \rightarrow \infty$. Contradiction!

We conclude that on the right side of the formula (2) differentiation under the integral sign is *not* legitimate. The cautionary tale illustrates the pitfalls of blind computation without proper justification.